

SEMESTER S5

COMPUTATIONAL STATISTICS

Course Code	PCCBT503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. In this course, students will explore the fundamental principles and techniques of computational statistics, emphasizing the application of statistical methods in data analysis and interpretation.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Multivariate Normal Distribution: Multivariate Normal Distribution Functions-Estimation of parameters., Conditional Distribution and its relation to regression model. Multivariate Regression: Assumptions of Multivariate Regression Models-Parameter estimation, Multivariate Analysis of variance and covariance.	9
2	Discriminant Analysis: Statistical background, linear discriminant function analysis, Estimating linear discriminant functions and their properties. Principal Component Analysis: Principal components, Algorithm for conducting principal component analysis, deciding on how many principal components to retain, H-plot.	9
3	Factor Analysis: Factor analysis model, extracting common factors, determining number of factors, Transformation of factor analysis solutions, Factor scores.	9

4	Cluster Analysis: Introduction, Types of clustering, Correlations and distances, clustering by partitioning methods, hierarchical clustering, overlapping clustering, K-Means Clustering-Profiling and Interpreting Clusters	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE)

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the techniques for the estimation of parameters in Multivariate Normal Distributions.	K3
CO2	Apply Multivariate Regression Models.	K3
CO3	Implement statistical analysis on the data.	K3
CO4	Implement the algorithm for conducting Principal Component Analysis.	K3
CO5	Master Factor Analysis Techniques.	K3
CO6	Apply Cluster Analysis Methods.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3	3	3							3
CO2	3	3	3	3	3							3
CO3	3	3	3	3	3							3
CO4	3	3	3	3	3							3
CO5	3	3	3	3	3							3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Regression Diagnostics, Identifying Influential Data and Sources of Collinearity	D.A. Belsey, E. Kuh and R.E. Welsch	Wiley	1980
2	Applied Linear Regression Models	J. Neter, W. Wasserman and M.H. Kutner.	McGraw-Hill/Irwin	2004 (4th Edition)
3	The Foundations of Factor Analysis	A.S. Mulaik.	McGraw-Hill	2009 (2nd Edition)
4	Introduction to Linear Regression Analysis	D.C. Montgomery and E.A. Peck	Wiley	2012 (5th Edition)
5	Cluster Analysis for Applications	M.R. Anderberg	Academic Press	1973
6	Multivariate Statistical Analysis	D.F. Morrison.	Duxbury Press	2005 (4th Edition)
7	Python for Data Analysis	Wes McKinney	O'Reilly Media	2022 (3rd Edition)

Video Links (NPTEL, SWAYAM...)	
Module No.	Link ID
1	https://archive.nptel.ac.in/courses/111/104/111104024/
2	https://onlinecourses.nptel.ac.in/noc24_ee118/preview
3	https://archive.nptel.ac.in/courses/106/106/106106139/
4	https://onlinecourses.nptel.ac.in/noc24_mg68/preview